

12 (**long**) The strain energy stored in a linear elastic material of Young's modulus E under a tensile stress σ and strain ε is given by $\sigma\varepsilon/2$, per unit volume.

(a) A sample of this material of length L and square cross-section $b \times b$, is loaded in tension to a stress $\sigma_{\max}$. Show that the elastic energy stored in the sample is equal to $b^2 L \sigma_{\max}^2 / 2E$. [3]

(b) An identical sample is loaded in pure bending with a moment M , as shown in Fig. 9, with a maximum stress at the surface equal to $\sigma_{\max}$. Use the Structures Databook to find the variation in stress $\sigma(y)$ and strain $\varepsilon(y)$, in terms of $\sigma_{\max}$ and the distance y from the neutral axis. Hence show that the elastic energy stored in pure bending is smaller than that in tension by a factor of 3, and explain qualitatively why this is the case. [10]

(c) Figure 10 shows two cross-sections of external dimensions $B \times B$. One section is solid, with an inner shaded region of dimensions $\alpha B \times \alpha B$. The second cross-section is hollow with internal dimensions $\alpha B \times \alpha B$. Samples of both sections, of length L , are loaded in pure bending until the maximum stress is equal to the elastic limit σ_y , as shown in Fig. 10. Use the result in (b) to calculate the elastic energy stored:

- (i) in the sample with the solid cross-section;
- (ii) in the material within the shaded region of the solid sample.

Use these results to show that the elastic energy stored in the hollow cross-section is equal to $(1 - \alpha^4) B^2 L \sigma_y^2 / 6E$. [9]

(d) Find the dimensions of a solid square cross-section of equal mass per unit length as the hollow cross-section in Fig. 10. Use the result in (b) to find the elastic energy stored in a sample of length L with this cross-section. What is the ratio of the elastic energy stored in the hollow cross-section, to the energy stored in the solid section of equal mass per unit length? What is the maximum possible value of this ratio as α varies? [8]

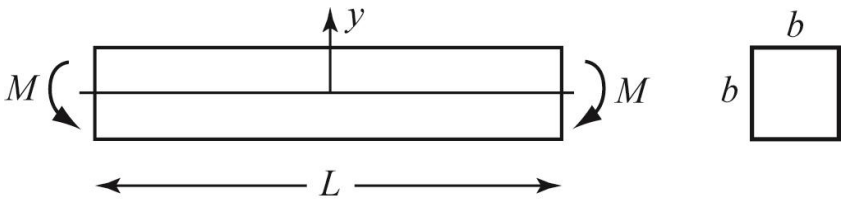


Fig. 9

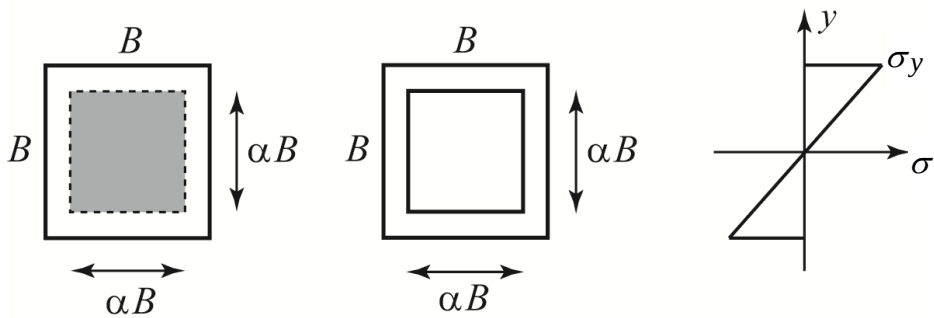


Fig. 10

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Paper 2

Numerical Answers

Section A (Structures)

2. $F_X = Mg \left(1 + \frac{\sqrt{3}}{2}\right), S_X = \frac{Mg}{2}, M_X = \frac{MgR}{2}$
3. $I = \frac{wh^3}{36}$
5. (a) AC: 20.6 kN; BC –20.0 kN, CD 10.2 kN; BD –10.7 kN.
(b) 91 mm downwards
6. (a) 116 MN m²
(b) $s \leq 0.17$ m.

Section B (Materials)

8. (b) $a \approx 2.87 \times 10^{-10}$ m, $R \approx 1.24 \times 10^{-10}$ m
11. (b) (i) $a_{\text{crit}} \approx 54.9$ mm, total crack length $2a_{\text{crit}} \approx 109.8$ mm
(iii) $n = 4$ and $A = 1.6 \times 10^{-11} \frac{\text{m/cycle}}{(\text{MPa}\sqrt{\text{m}})^4}$.
(iv) $N_f \approx 229 \times 10^3$ cycles